

## DNIPROPETROVSK, Ukraine

WMO: 345040

Lat: 48.3600N

Lon: 35.0850E

Elev: 469

StdP: 14.45

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.2       | 5.3 | -5.5                          | 4.2 | 0.5  | 0.1 | 5.6 | 6.2  | 28.1                  | 11.2 | 26.0 | 14.3 | 9.3                   | 30   | 0.585 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.6                   | 93.4            | 69.6 | 89.9 | 68.7 | 86.7 | 67.8 | 72.7                | 86.8 | 71.0 | 84.6 | 69.6 | 82.6 | 9.9                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.3                            | 105.6 | 78.4 | 66.5 | 99.4 | 76.3 | 65.1 | 94.4 | 74.7 | 36.6          | 86.7 | 35.2 | 84.9 | 33.9 | 82.6 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |       |            |       |            |       |       |       |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|-------|------------|-------|------------|-------|-------|-------|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |       | n=20 years |       | n=50 years |       |       |       |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max   | Min        | Max   | Min        | Max   |       |       |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |       |
| (4)               | 23.4 | 20.8 | 18.6                       | DB  | -5.1               | 98.3 | 7.1                                                | 4.2 | -10.2      | 101.3 | -14.4      | 103.8 | -18.4      | 106.1 | -23.5 | 109.2 |
| (5)               |      |      |                            | WB  | -5.6               | 76.4 | 7.0                                                | 2.5 | -10.7      | 78.2  | -14.7      | 79.6  | -18.7      | 81.0  | -23.7 | 82.8  |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 49.7   | 25.0  | 27.3  | 36.7  | 51.0  | 62.4  | 69.3  | 73.4  | 72.4  | 61.9  | 49.0  | 37.7  | 28.8 |      |
|                                                                                  | DBStd                                                                            | 19.18  | 10.01 | 9.86  | 7.94  | 7.56  | 7.83  | 6.52  | 5.86  | 6.79  | 7.76  | 8.35  | 8.37  | 9.37 |      |
|                                                                                  | HDD50                                                                            | 3069   | 776   | 635   | 414   | 77    | 6     | 0     | 0     | 0     | 6     | 121   | 377   | 657  |      |
|                                                                                  | HDD65                                                                            | 6372   | 1241  | 1055  | 876   | 421   | 144   | 31    | 5     | 13    | 147   | 498   | 819   | 1122 |      |
|                                                                                  | CDD50                                                                            | 2962   | 0     | 0     | 3     | 108   | 391   | 580   | 724   | 694   | 363   | 90    | 9     | 0    |      |
|                                                                                  | CDD65                                                                            | 787    | 0     | 0     | 0     | 1     | 64    | 161   | 263   | 241   | 54    | 3     | 0     | 0    |      |
|                                                                                  | CDH74                                                                            | 7510   | 0     | 0     | 0     | 33    | 638   | 1456  | 2475  | 2395  | 490   | 23    | 0     | 0    |      |
|                                                                                  | CDH80                                                                            | 2965   | 0     | 0     | 0     | 6     | 189   | 516   | 1031  | 1074  | 146   | 3     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 9.2    | 9.9   | 10.4  | 10.6  | 9.3   | 8.3   | 8.4   | 8.1   | 8.7   | 8.8   | 8.8   | 9.5   | 10.1 |      |
| Precipitation                                                                    | PrecAvg                                                                          | 20.0   | 1.6   | 1.5   | 1.4   | 1.5   | 1.7   | 2.3   | 2.0   | 1.9   | 1.4   | 1.4   | 1.7   | 1.9  |      |
|                                                                                  | PrecMax                                                                          | 33.5   | 5.3   | 3.0   | 3.5   | 3.5   | 5.5   | 6.1   | 5.1   | 8.5   | 4.7   | 5.0   | 4.9   | 5.0  |      |
|                                                                                  | PrecMin                                                                          | 13.1   | 0.2   | 0.1   | 0.2   | 0.0   | 0.0   | 0.3   | 0.0   | 0.1   | 0.1   | 0.1   | 0.0   | 0.2  |      |
|                                                                                  | PrecStd                                                                          | 4.6    | 1.1   | 0.7   | 0.8   | 0.9   | 1.1   | 1.3   | 1.4   | 1.6   | 1.1   | 1.1   | 1.1   | 1.1  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 46.1  | 49.9  | 63.0  | 78.1  | 89.8  | 93.0  | 97.7  | 100.1 | 89.7  | 77.2  | 61.2  | 50.4 |      |
|                                                                                  |                                                                                  | MCWB   | 44.4  | 46.0  | 49.5  | 58.5  | 66.1  | 69.4  | 71.7  | 69.7  | 65.7  | 61.1  | 54.7  | 48.3 |      |
|                                                                                  | 2%                                                                               | DB     | 42.2  | 44.8  | 57.2  | 72.1  | 84.7  | 89.3  | 93.1  | 95.0  | 84.1  | 70.3  | 56.6  | 46.7 |      |
|                                                                                  |                                                                                  | MCWB   | 41.2  | 42.3  | 47.2  | 55.7  | 64.7  | 68.5  | 70.7  | 69.0  | 63.5  | 58.8  | 52.7  | 45.5 |      |
|                                                                                  | 5%                                                                               | DB     | 39.3  | 41.2  | 52.2  | 68.3  | 81.0  | 86.0  | 89.6  | 89.9  | 79.3  | 65.8  | 52.5  | 43.3 |      |
|                                                                                  |                                                                                  | MCWB   | 38.4  | 39.4  | 44.8  | 54.3  | 63.6  | 67.9  | 69.7  | 68.3  | 62.3  | 56.9  | 50.0  | 42.3 |      |
|                                                                                  | 10%                                                                              | DB     | 35.9  | 38.4  | 48.3  | 64.5  | 76.9  | 82.5  | 86.1  | 86.0  | 75.2  | 60.8  | 49.0  | 40.9 |      |
|                                                                                  |                                                                                  | MCWB   | 35.1  | 36.8  | 42.9  | 52.0  | 61.8  | 66.7  | 68.9  | 67.3  | 60.7  | 54.9  | 47.1  | 40.1 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 44.6  | 47.3  | 51.8  | 60.3  | 69.7  | 73.5  | 75.6  | 73.8  | 69.0  | 64.1  | 56.4 | 49.1 |
|                                                                                  |                                                                                  |        | MCDB  | 45.4  | 49.0  | 60.4  | 74.2  | 82.4  | 86.4  | 88.6  | 89.3  | 82.8  | 72.3  | 60.0 | 50.1 |
| 2%                                                                               |                                                                                  | WB     | 41.1  | 42.7  | 48.5  | 57.5  | 66.9  | 71.0  | 73.4  | 71.8  | 66.2  | 60.6  | 53.4  | 45.6 |      |
|                                                                                  |                                                                                  | MCDB   | 41.6  | 44.2  | 55.2  | 68.9  | 80.3  | 83.8  | 87.4  | 87.2  | 78.2  | 66.8  | 56.3  | 46.7 |      |
| 5%                                                                               |                                                                                  | WB     | 38.2  | 39.5  | 45.8  | 55.4  | 64.9  | 69.5  | 71.7  | 70.0  | 64.1  | 57.9  | 50.5  | 42.5 |      |
|                                                                                  |                                                                                  | MCDB   | 38.9  | 40.8  | 50.9  | 65.8  | 77.6  | 82.0  | 85.1  | 84.8  | 74.9  | 64.2  | 52.3  | 43.2 |      |
| 10%                                                                              |                                                                                  | WB     | 35.3  | 36.9  | 43.2  | 53.5  | 63.0  | 67.7  | 70.1  | 68.4  | 62.3  | 55.4  | 47.2  | 39.9 |      |
|                                                                                  |                                                                                  | MCDB   | 35.9  | 38.0  | 47.2  | 62.6  | 75.2  | 79.5  | 82.7  | 82.3  | 72.1  | 60.5  | 48.7  | 40.5 |      |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB                                                                            | MDBR   | 9.1   | 10.9  | 14.1  | 18.7  | 20.6  | 19.9  | 20.6  | 21.4  | 19.8  | 15.8  | 10.4  | 8.5  |      |
|                                                                                  |                                                                                  | MCDBR  | 9.0   | 12.4  | 21.5  | 24.5  | 25.3  | 24.2  | 24.9  | 26.6  | 25.4  | 22.2  | 14.7  | 9.9  |      |
|                                                                                  |                                                                                  | MCWBR  | 8.1   | 10.0  | 13.4  | 12.3  | 11.6  | 10.1  | 9.5   | 9.4   | 11.3  | 12.4  | 10.8  | 9.1  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 8.5   | 11.5  | 19.4  | 21.9  | 22.8  | 21.2  | 21.9  | 22.9  | 21.9  | 19.5  | 13.4  | 9.6  |      |
|                                                                                  |                                                                                  | MCWBR  | 8.0   | 9.8   | 13.1  | 11.6  | 11.2  | 10.1  | 9.8   | 9.5   | 10.7  | 12.2  | 10.6  | 9.1  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.328  | 0.356 | 0.386 | 0.449 | 0.415 | 0.412 | 0.437 | 0.436 | 0.409 | 0.370 | 0.352 | 0.331 |      |      |
|                                                                                  | taud                                                                             | 2.214  | 2.166 | 2.211 | 2.082 | 2.270 | 2.298 | 2.233 | 2.227 | 2.269 | 2.358 | 2.315 | 2.274 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 225    | 246   | 259   | 253   | 269   | 270   | 261   | 255   | 249   | 241   | 216   | 208   |      |      |
|                                                                                  | Edh at Noon                                                                      | 25     | 33    | 38    | 48    | 41    | 40    | 42    | 41    | 36    | 28    | 23    | 22    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 296    | 573   | 941   | 1431  | 1854  | 1978  | 1964  | 1766  | 1223  | 731   | 327   | 212   |      |      |
|                                                                                  | RadStd                                                                           | 57     | 112   | 124   | 131   | 157   | 169   | 120   | 110   | 148   | 109   | 55    | 42    |      |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only           | +1.42 | N/A     | N/A    | +2.02   | N/A   | N/A   | -315        | -392  | +211  | +137  |
| (47) | Regional (9 neighbors) | +1.40 | N/A     | N/A    | +1.94   | N/A   | N/A   | -259        | -371  | +227  | +144  |

Nomenclature: See separate page